Summary Test Maths

This covers some remaining parts from sections of Algebra (A4) from the edexcel specification.

Topics tested (Use Maths Genie to learn if needed):

1. Expanding and Factorising (Q1 - Q5)
2. Expanding and Factorising Quadratics (Q6 - Q8)
3. Factorising Harder Quadratics (Q9 - Q13)
4. Surds (Q14 - Q19)

Q1.

(a) Expand $7(2x + 7)$

(b) Factorise $3y + 12$

Q2.

(a) Factorise fully $12m + 8m^2$

(b) Solve $3(n - 5) = 27$

Q3.

(a) Factorise fully $5a^2b + 15ab^2$

(b) Solve $6(c - 8) = 42$

Q4.

(a) Factorise fully $30x^3 + 12x$

Q5.

(a) Factorise fully $18a^2bc + 30abc^2$

Q6.

(a) Expand and simplify $(2x + 3)(3x - 1)$

(b) Factorise $x^2 + 10x + 25$

Q7. (Hint part (b) - difference of 2 squares)

(a) Expand and simplify $(5h + 2)(h + 4)$

(b) Factorise $x^2 - 49$

Q8.

(a) Expand and simplify $(7x + 1)(x + 5)$

(b) Factorise $y^2 + 13y + 30$

Q9.

Factorise $2x^2 + 5x + 2$

Q10.

Solve $4x^2 - 19x - 5 = 0$

Q11.

Factorise fully $2x^2 - 98$

Q12.

Factorise $9x^2 - 3x - 20$

Q13.

Factorise $2x^2 + 13xy + 15y^2$

Q14.

Write $\sqrt{48}$ in the form $k\sqrt{3}$, where k is an integer.

Q15.

Expand and Simplify $(2 + \sqrt{3})(2 - \sqrt{3})$

Q16.

Write $(3 - \sqrt{2})^2$ in the form $a + b\sqrt{2}$, where a and b are integers.

Q17.

Rationalise the denominator $\frac{6}{\sqrt{3}}$

Q18.

Simplify $\frac{(3 + \sqrt{6})}{\sqrt{3}}$

Q19.

Show that $\frac{3\sqrt{3} + 3}{3 + \sqrt{3}}$ can be written as $\sqrt{3}$

Answers:

A1.

(a) Expand $7(2x + 7)$

$14x + 49$

(1)

(b) Factorise $3y + 12$

$3(y + 4)$

A2.

(a) Factorise fully $12m + 8m^2$

$4m(3 + 2m)$

(2)

(b) Solve $3(n - 5) = 27$

$3n - 15 = 27$

$3n = 42$

$n =$ 14

A3.

(a) Factorise fully $5a^2b + 15ab^2$

$5ab(a + 3b)$

(2)

(b) Solve $6(c - 8) = 42$

$6c - 48 = 42$

$6c = 90$

$c = 15$

$c =$ 15

(2)

A4.

(a) Factorise fully $30x^3 + 12x$

$6x(5x^2 + 2)$

A5.

(a) Factorise fully $18a^2bc + 30abc^2$

$6abc(3a + 5c)$

A6.

(a) Expand and simplify $(2x + 3)(3x - 1)$

$6x^2 - 2x + 9x - 3$

$6x^2 + 7x - 3$

(2)

(b) Factorise $x^2 + 10x + 25$

25

1 25

5 5

$(x + 5)(x + 5)$

A7.

(a) Expand and simplify $(5h + 2)(h + 4)$

$5h^2 + 20h + 2h + 8$

$5h^2 + 22h + 8$

(2)

(b) Factorise $x^2 - 49$

$(x + 7)(x - 7)$

(1)

A8.

(a) Expand and simplify $(7x + 1)(x + 5)$

$7x^2 + 35x + x + 5$

$7x^2 + 36x + 5$

(2)

(b) Factorise $y^2 + 13y + 30$

1	30
2	15
3	10
5	6

$(y + 3)(y + 10)$

(2)

A9.

Factorise $\overset{A}{2}x^2 + 5x + \overset{C}{2}$

$A \times C = 4$

1×4

2×2

$$\frac{(2x + 1)(2x + 4)}{2}$$

$(2x + 1)(x + 2)$

$(2x+1)(x+2)$

A10.

Solve $4x^2 - 19x - 5 = 0$

$AC = 20$

1	20
2	10
4	5

$$\frac{(4x+1)(4x-20)}{4}$$

$$(4x+1)(x-5) = 0$$

$$x = -\frac{1}{4} \quad x = 5$$

$$x = -\frac{1}{4} \quad x = 5$$

A11.

Factorise fully $2x^2 - 98$

$$2(x^2 - 49)$$

$$2(x+7)(x-7)$$

$$2(x+7)(x-7)$$

A12.

Factorise $9x^2 - 3x - 20$

$$\frac{(9x+12)(9x-15)}{9}$$

$$(3x+4)(3x-5)$$

180	
1	180
2	90
3	60
4	45
5	36
6	30
9	20
10	18
12	15

$(3x+4)(3x-5)$

A13.

Factorise $2x^2 + 13xy + 15y^2$

$$\frac{(2x+3y)(2x+10y)}{2}$$

$$(2x+3y)(x+5y)$$

30	
1	30
2	15
3	10
5	6

$(2x+3y)(x+5y)$

A14.

Write $\sqrt{48}$ in the form $k\sqrt{3}$, where k is an integer.

$\sqrt{16}\sqrt{3}$

$4\sqrt{3}$

$4\sqrt{3}$

A15.

Expand and Simplify $(2+\sqrt{3})(2-\sqrt{3})$

$4 - 2\sqrt{3} + 2\sqrt{3} - 3$

1

1

A16.

Write $(3-\sqrt{2})^2$ in the form $a + b\sqrt{2}$, where a and b are integers.

$(3-\sqrt{2})(3-\sqrt{2})$

$9 - 3\sqrt{2} - 3\sqrt{2} + 2$

$11 - 6\sqrt{2}$

$11 - 6\sqrt{2}$

A17.

Rationalise the denominator $\frac{6}{\sqrt{3}} \frac{\times\sqrt{3}}{\times\sqrt{3}}$

$\frac{6\sqrt{3}}{3}$

$2\sqrt{3}$

$2\sqrt{3}$

A18.

Simplify $\frac{(3+\sqrt{6})}{\sqrt{3}} \begin{matrix} \times\sqrt{3} \\ \times\sqrt{3} \end{matrix}$

$\sqrt{18} = \sqrt{9}\sqrt{2}$
$= 3\sqrt{2}$

$$\frac{3\sqrt{3} + \sqrt{18}}{3}$$

$$\frac{3\sqrt{3} + 3\sqrt{2}}{3}$$

$$\cancel{\frac{6\sqrt{2}}{3}} \qquad \underline{\underline{\sqrt{3} + \sqrt{2}}}$$

A19.

Show that $\frac{3\sqrt{3}+3}{3+\sqrt{3}}$ can be written as $\sqrt{3}$

$$\frac{(3\sqrt{3}+3)(3-\sqrt{3})}{(3+\sqrt{3})(3-\sqrt{3})}$$

$$\frac{9\sqrt{3} - 3(3) + 9 - 3\sqrt{3}}{9 - 3\sqrt{3} + 3\sqrt{3} - 3}$$

$$\frac{6\sqrt{3}}{6}$$

$$\underline{\underline{\sqrt{3}}}$$

AQA GCSE Combined Science – Chemistry
Topic 5.1: Atomic Structure and the Periodic Table
Summary Test

1. What is an atom? (1 mark)

2. What is an element? (1 mark)

3. What is a compound? (1 mark)

4. Complete the table of subatomic particles. (3 marks)

Proton – charge? relative mass?

Neutron – charge? relative mass?

Electron – charge? relative mass?

5. Which particle has a negative charge? (1 mark)

A) Proton B) Neutron C) Electron

6. Name the separation technique used to separate: (3 marks)

a) Sand from water

b) Water from salt water

c) Different dyes in ink

7. What is a mixture? (1 mark)

8. Explain the difference between a compound and a mixture. (2 marks)

9. Describe the plum pudding model of the atom. (2 marks)

10. The gold foil experiment led to a new model of the atom.

a) Who carried out this experiment? (1 mark)

b) What did the experiment show about the structure of the atom? (2 marks)

11. What is an isotope? (2 marks)

12. Write the electron configuration for: (3 marks)

a) Oxygen

b) Sodium

c) Calcium

13. Describe how you would obtain pure copper sulfate crystals from copper sulfate solution. (3 marks)

14. Explain why elements in the same group have similar chemical properties. (2 marks)

15. An atom has atomic number 12 and mass number 24. Complete: (3 marks)

Protons = ? Neutrons = ? Electrons = ?

16. Describe Rutherford's alpha scattering experiment and explain how the results led to the nuclear model. (4 marks)

17. Explain why fractional distillation can separate liquids in a mixture. (3 marks)

18. In chromatography a dye travels 4 cm and the solvent travels 8 cm. Calculate the Rf value. (2 marks)

19. Explain why isotopes of the same element have the same chemical properties. (2 marks)

20. Describe how the atomic model developed from Dalton to the modern model. (6 marks)

Mark Scheme

1. Smallest part of an element that can exist.

2. Substance made of only one type of atom.

3. Two or more different elements chemically bonded together.

4. Proton: +1, mass 1 | Neutron: 0, mass 1 | Electron: –1, mass 1/1836.

5. C) Electron.

6. a) Filtration b) Simple distillation c) Chromatography.

7. Two or more substances not chemically bonded together.

8. Compound: elements chemically bonded, new properties. Mixture: not chemically bonded and can be separated physically.

9. Atom is a sphere of positive charge with electrons embedded throughout.

10a. Ernest Rutherford.

10b. Atom mostly empty space; small dense positively charged nucleus.

11. Atoms of the same element with the same number of protons but different numbers of neutrons.

12. Oxygen: 2,6 | Sodium: 2,8,1 | Calcium: 2,8,8,2.

13. Heat solution to evaporate some solvent, allow to cool, crystals form and can be filtered/dried.

14. Same number of outer shell electrons.

15. Protons = 12, Electrons = 12, Neutrons = 12.

16. Alpha particles fired at thin gold foil; most passed through, some deflected, few bounced back; conclusion: atom mostly empty space with small dense nucleus.

17. Liquids have different boiling points; lowest boiling point evaporates first and condenses separately.

18. Rf = 4 ÷ 8 = 0.5.

19. Same number of electrons and same electronic structure.

20. Dalton: solid sphere → Thomson: electron & plum pudding → Rutherford: nucleus (gold foil) → Bohr: electron shells → Chadwick: neutron.

Revision in flashcards

Topic 1 - Atomic structure and the periodic table

5.1 Atomic structure and the periodic table
5.1.1 A simple model of the atom, symbols, relative atomic mass, electronic charge and isotopes

5.1.1.1 Atoms, elements and compounds

Q1. What is an atom?
A1. An atom is the smallest part of an element. They are the building blocks of all substances.

Q2. What is an element?
A2. A substance made of only one type of atom. There are about 100 elements and can be found in the periodic table.

Q3. How can elements be represented?
A3. They can be represented via a chemical symbol. For example the chemical symbol (Na) represents **one** atom of the element Sodium.

Q4. What is a compound?
A4. A substance formed when 2 or more different elements chemically bond together. For example water is 2 hydrogen and one oxygen atom chemically bonded together.

Q5. Compounds can also be represented with chemical symbols by combining the elements that make up the compound. If water is 2 hydrogen and one oxygen atom what would its chemical symbol be?
A5. H_2O

Q6. What do chemical reactions involve?
A6. The formation of one or more new substances, and often a detectable energy change.

Q7. How can compounds be separated into their elements?
A7. Through chemical reactions.

Q8. How do compounds mainly differ from the elements that make them up?
A8. Compounds have different unique properties from the elements they contain. E.g - the element sodium is a reactive metal and chlorine is a toxic gas but its compound sodium chloride is just table salt.

5.1.1.2 Mixtures

Q9. What is a mixture?
A9. Two or more elements or compounds **not** chemically combined together.

Q10. How is a mixture different from a compound?
A10.

- The chemical properties of each substance in the mixture are unchanged (keep original properties).
- A mixture can be separated with physical processes, unlike compounds that can only be separated via chemical reactions.

Q11. What is a physical process in the context of separating mixtures?
A11. A method of separation that does not involve chemical reactions and no new substances are made.

Q12. Name the physical process that can be used to separate mixtures.
A12.

- Filtration
- Crystallisation
- Simple distillation
- Fractional distillation
- Chromatography

Q13. Briefly describe and explain each physical process used for separating mixtures.
A13.
Filtration - Separates insoluble (a solid that doesn't dissolve in a liquid) solid from a liquid e.g sand + water.

- Mixture poured through filter paper in a funnel.
- Liquid passes through.
- Solid stays on the filter paper.

Crystallisation - Separates a soluble (solid that does dissolve in liquid) solid from a liquid e.g salt from salt water.

- Heat solution to evaporate the solvent (usually the liquid that dissolved the soluble solid).
- This leaves behind the separated solid in a crystal form. (you have to be careful that heating the liquid doesn't effect the thing we are trying to crystalise)

Simple distillation - Separates a liquid from a solid if we want to keep the liquid e.g water from salt water. (crystalisation does the same thing but the liquid is lost through evaporation)

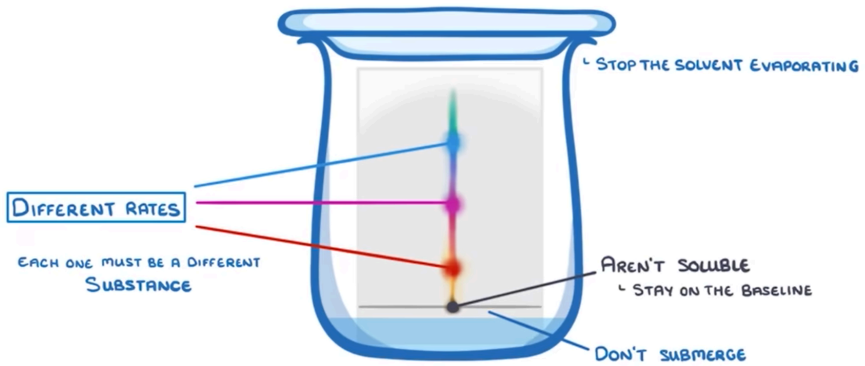
- Evaporate the liquid by heating.
- Vapour travels into a condenser cooling it.
- Vapour cools and becomes liquid (distillate)

Fractional distillation - Separate mixtures of liquids (they MUST have different boiling points for this method to work) e.g - ethanol and water.

- Mixture heated.
- The liquid with the lowest boiling point evaporates.
- The vapour travels up the fractionating column.
- The column allows vapours to cool and condense repeatedly, helping separate liquids with similar boiling points.
- It condenses and is collected as a liquid.

Chromatography (specifically **paper** chromatography) - Separates mixtures of soluble substances such as inks and food colorings based on their different solubilities in a solvent.

- Take a piece of chromatography paper and draw a line in **pencil** near the bottom. This is called the baseline.
- Add your sample of ink you suspect to contain a mixture of substances to the pencil line.
- Get your solvent (water or ethanol).
- Add the chromatography paper to the solvent making sure you don't submerge it more than the pencil line.
- Wait giving the solvent time to go up the paper dissolving your sample.
- Leave your paper to dry.
- You are left with a chromatogram.



Q14. Explain how paper chromatography works?

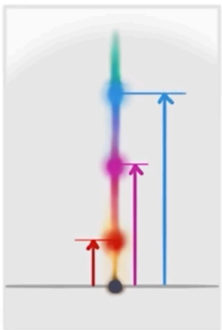
A14.

- As the solvent goes up the paper the different substances that make up our sample will dissolve in the solvent at different rates.
- Our substances dissolve in the solvent. Move a little bit (this is called the mobile phase since they are moving) and then, once again binding to the paper and staying still (called the stationary phase). Then they will again move a little bit, then stop moving again and so on. This happens over and over.
- Since more soluble substances will spend more time in the mobile phase than less soluble ones which will spend more time in the stationary phase. This makes the more soluble substance move 'faster' and higher up the paper.
- This separates the different substances since they have different solubilities hence spend more or less time in the mobile phase.

- If there are any insoluble substances in the sample they will not dissolve and won't move up the paper staying at the baseline.

Q15. How can we use the chromatogram to identify substances?
A15.

- How far each substance moves up the paper is dependent on the properties of the substance. Hence will be the same everytime anyone does the experiment.
- So you can measure the length of how much a substance moves up the paper and compare it to a data book which has the lengths and corresponding substance and use that to identify your substance.



Q16. What is a better way to identify the substance in chromatography than just looking at the length?
A17.

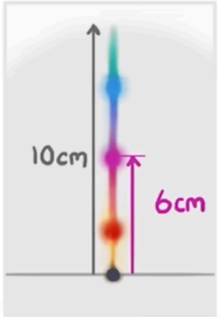
- How far the substance moves up the paper is dependent on how long you leave the experiment for. (Leaving it longer means the substance moves up more).
- So instead of comparing just the length we calculate the ratio called the R_f value.

Q18. How is the R_f value calculated?
A18.

$$\text{'}R_f \text{ VALUE'} = \frac{\text{DISTANCE TRAVELLED BY THE SUBSTANCE}}{\text{DISTANCE TRAVELLED BY THE SOLVENT}}$$

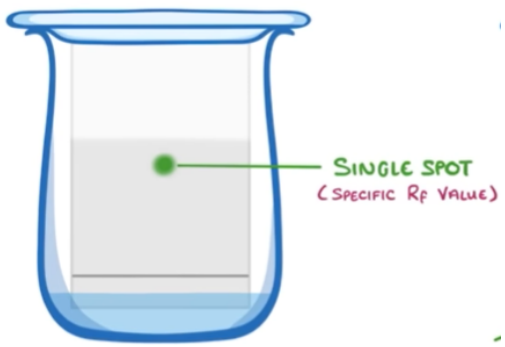
Q19. Why is the R_f value more reliable?
A19.
It is more reliable since it's not affected by time.
If the experiment is left for longer, indeed the substance will move higher up the paper giving us a different length but so will the solvent. Hence giving us the same ratio (R_f value) no matter how long we left the experiment for.
You can the use a data book of R_f values and compare to identify your substance.

Q20. If a substance moves up 6cm from the baseline and the solvent moves up 10cm. What is the R_f value?

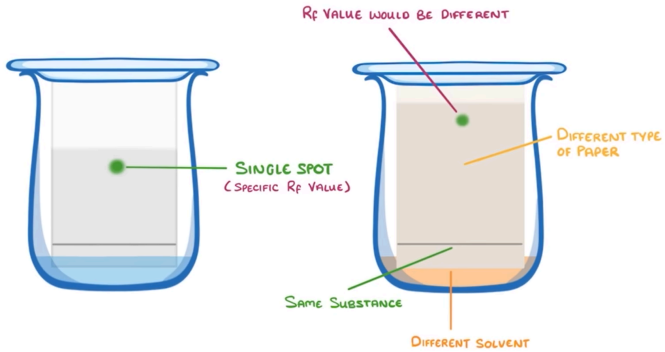


A20. $R_f = 6 / 10 = 0.6$

Q21. What will happen if we use paper chromatography on a pure substance that has nothing else mixed into it ?
A21. If the substance is soluble it will still move up the paper but won't separate out. Leaving a single spot on the paper. You can calculate a specific R_f value for this.



Q22. What happens if you put the same pure substance but in a different solvent or use a different type of paper.
A22. Since the substance might be more or less soluble with that solvent or more or less clinging to the paper. It will spend less or more time in the mobile and stationary phase. So you will still get a single spot since there is nothing to separate but it will be in a different place along the paper.



Because of this when looking up R_f values you also need to look up the solvent and paper used.

5.1.1.3 The development of the model of the atom

Q23. What was the initial idea of what atoms where and who proposed it?
A23. John Dalton - thought atoms to be tiny spheres that could not be divided.

Q24. What did JJ Thomson discover and how did this change our understanding of the atom?
A24. He discovered electrons, which lead to the plum pudding model.

Q25. What is the plum pudding model?
A25.

- Atom = sphere of positive charge.
- Electrons embedded throughout.
- It was called plum pudding because electrons were like plums in a pudding.

Q26. What experiment and by who lead to the disproving of the plum pudding model.
A26. Rutherford's - Alpha particle scattering experiment (also called gold foil experiment) led to the plum pudding model being disproved.